

# Autodesk® Scaleform®

## AMP

Scaleform AMP(Analyzer for Memory and Performance)

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: 3.03  
: 2013 4 10

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## Autodesk® Scaleform® 4.3

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Autodesk Scaleform

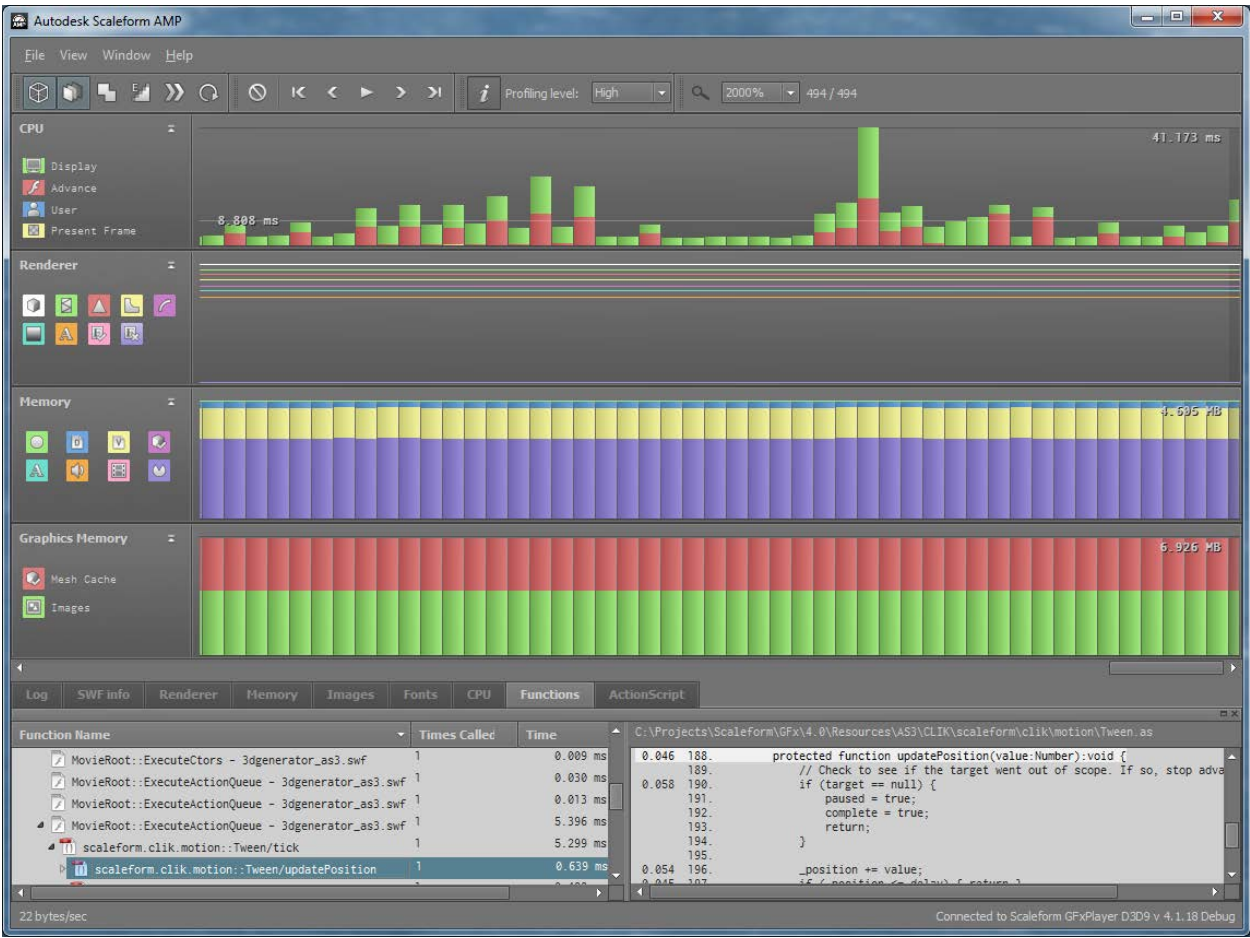
---

AMP  
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(301) 446-3200  
(301) 446-3199

|          |            |           |
|----------|------------|-----------|
| <b>1</b> |            | <b>1</b>  |
| 1.1      | AMP        | 2         |
| 1.2      | AMP        | 3         |
| <b>2</b> | <b>AMP</b> | <b>13</b> |
| 2.1      |            | 13        |
| 2.2      |            | 14        |
| 2.3      |            | 15        |
| 2.4      |            | 16        |
| 2.5      |            | 24        |
| 2.6      |            | 25        |
| <b>3</b> |            | <b>26</b> |
| <b>4</b> |            | <b>28</b> |
| <b>5</b> | <b>AMP</b> | <b>30</b> |
| 5.1      | Debug      | 30        |
| 5.2      |            | 30        |
| 5.3      |            | 31        |
| 5.4      |            | 32        |
| <b>6</b> | <b>AMP</b> | <b>33</b> |
|          | <b>FAQ</b> |           |
| <b>7</b> |            | <b>35</b> |

1

AMP™   Scaleform®   , Scaleform   CPU   ,   ,   .   , AMP   ActionScript



1: Scaleform AMP

Scaleform   AMP   .   ,   AMP   AMP

AMP    Scaleform 3.2

## 1.1 AMP

Scaleform (AMP )

Scaleform .    ActionScript    C++

Scaleform . ActionScript 2

Scaleform . (ActionScript 3

Scaleform .) , flash ,

ActionScript .

APM    AMP

Shipping AMP ,

AMP .

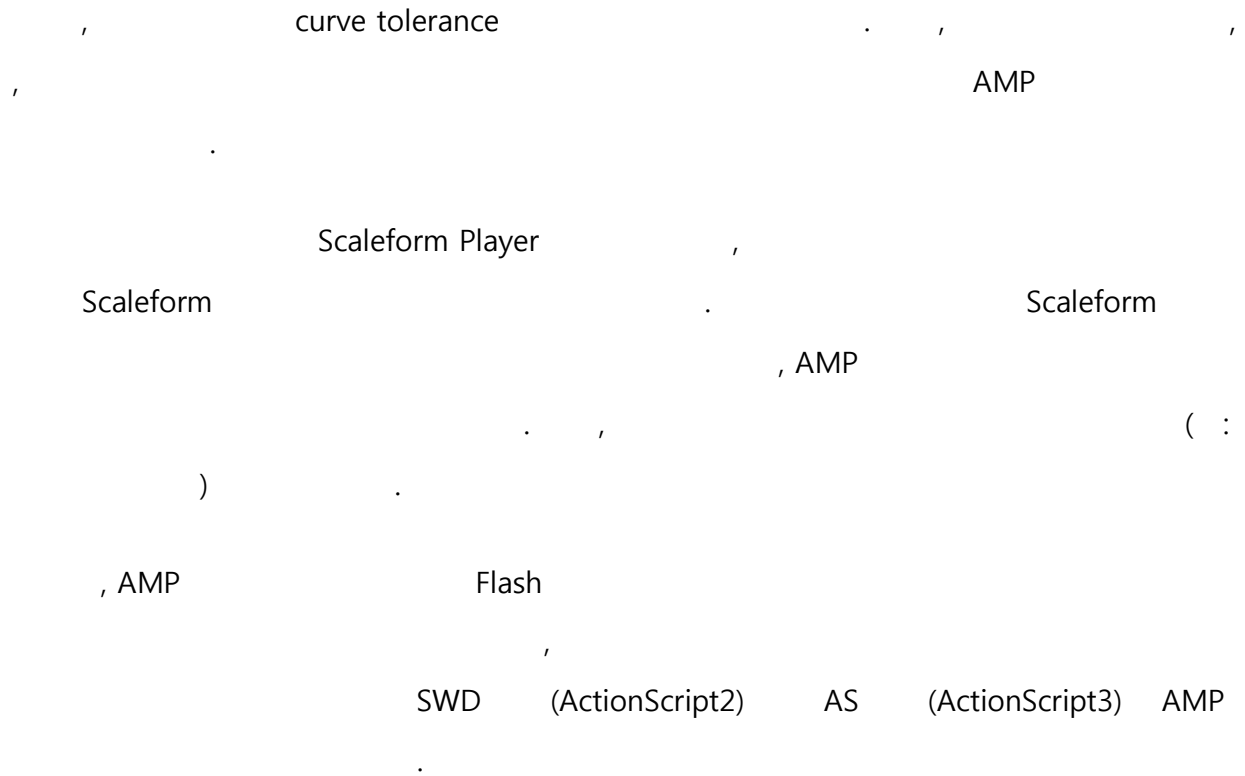
Movie .    SWF

view . , AMP

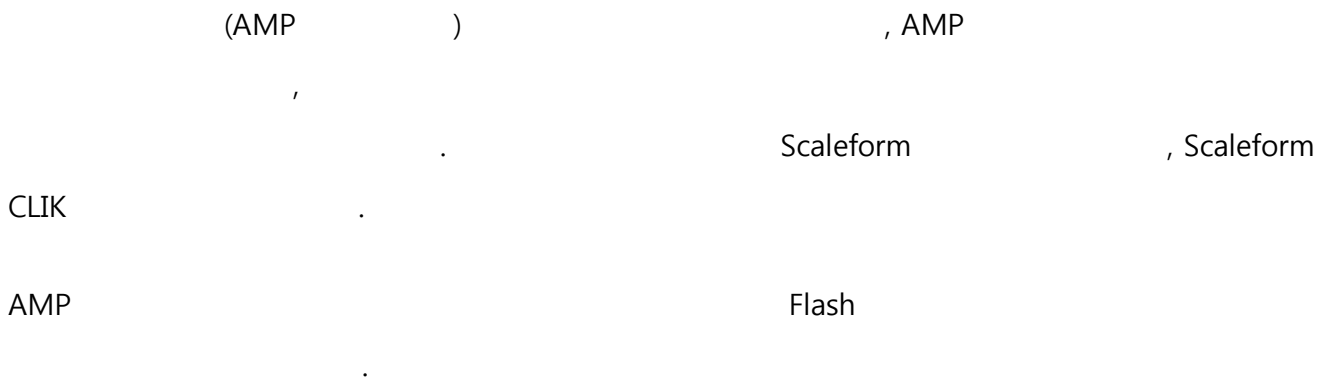
Movies    ActionScript .

AMP

flash movie , , ,



## 1.2 AMP



- **CPU graph:** ActionScript(Advance), Scaleform (Display), Direct Access API (User), (PresentFrame) CPU
- Scaleform CPU

(legend)

(toggle)

- **Rendering graph:** [Best Practices Guide](#)

(primitive),

)

Scaleform

- **Memory graph:** Scaleform

$$\left( \begin{array}{cccc} & & & \\ & & & \\ & & & \\ & & & \end{array} \right)$$

- **Graphics memory graph:** Scaleform

AMP

GPU

- **Log tab:** Log Scaleform

- **Performance Adviser:** Flash

. Performance Adviser

- o *Text Fields optimized for readability.*

11

11

11

- o *Ratio of draw primitives to number of meshes.*

(1.0 )

- o *Number of draw primitives.*



- o *Number of meshes.* ,
- o *Number of triangles.*
- o *Mesh cache thrashing.*
- o *Number of strokes.* " "
- o *Number of gradient fills.* " "
- o *Number of masks.* " "
- o *Number of filters.* " "
- o *Number of DirectAccess API calls.* Direct Access API ,
- o *Number of rasterized glyphs.* ,
- o *Number of font textures.* ,
- o *Number of font texture updates.* .
- o *Font cache thrashing.*
- o *Unused font texture percentage.* ,

- o *Used font cache area:*
- o *Total font cache area:*
- o *Total Memory:* Scaleform
- o *MovieData Memory:*
- o *Font Memory:*
- o *MovieView Memory:* Gfx::Movie  
. MovieView
- o *ActionScript Memory:* ActionScript
- o *Mesh Cache Memory:*
- o *Unused Mesh Cache Memory:*
- o *Image Memory:*
- o *Video Memory:* GPU Scaleform
- o *Sound Memory:*
- o *Number of images exceeding a certain size:*

AMP

AMP

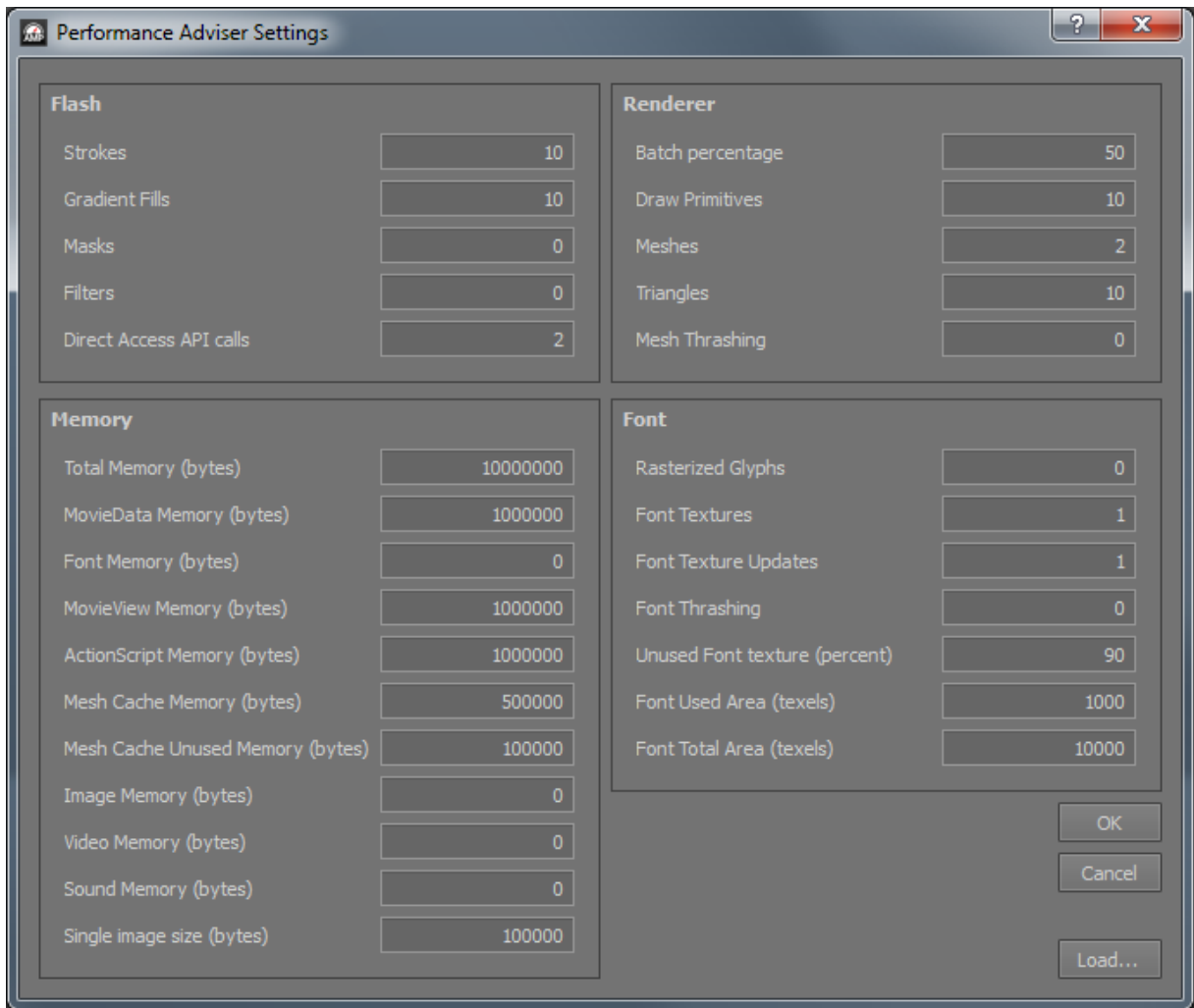
"adviser.xml"

XML

0

Performance Advisor

Settings



OK

AMP

XML

Adviser

"-adviser

<AdviserFilePath>"

AMP

Adviser

- **SWF info tab:**

Frame rate

- **Renderer tab:**

(

),

(

- **Memory tab:** Scaleform, Scaleform
- **Images tab:** Scaleform, AMP, (DXT).
- **Fonts tab:** SWF, Images
- **CPU tab:** Advance, Display, User
- **Functions tab:** . ActionScript C++, ActionScript VM
- **ActionScript tab:** Functions, C++
- **Source pane:** Functions, ActionScript, . AS3, AS, . SWF, AS3, .. AS2, ( ), " " . AS2

## Scaleform

. 1

SWF  
AMP  
AS3  
flash

- **File Menu:**

- o **Connection:**

- o **Disconnect:**

- o **Debug Info Paths:**

- (AS3 )

- Remove

Add

- o **Load Profile Frames:**

- o **Save Profile Frames:**

- "Load Profile Frames"

- o **Dump Memory Report:**

- "AmpMemReport.txt"

- o **Exit:**

- **View menu:**

- o **Graph Tooltips:**

- AMP



HW FSAA( ) None( )

- o  **Fast-forward:** flash  
SWF  
FPS

- o  **Restart:** flash

- **Frame control toolbar:**

- o  **Clear:**  
AMP

- o  **First Frame:**

- o  **Previous Frame:**

- o  **Pause:**

- o  **Next Frame:**

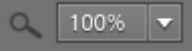
- o  **Last Frame:**

- **Profiling toolbar:**

o  **Detailed Profiling**( ): AMP

Memory

AMP Images 10

-  **Zoom toolbar:**

- **Status bar:**

Status Bar

AMP

Window

Reset UI

UI

AMP



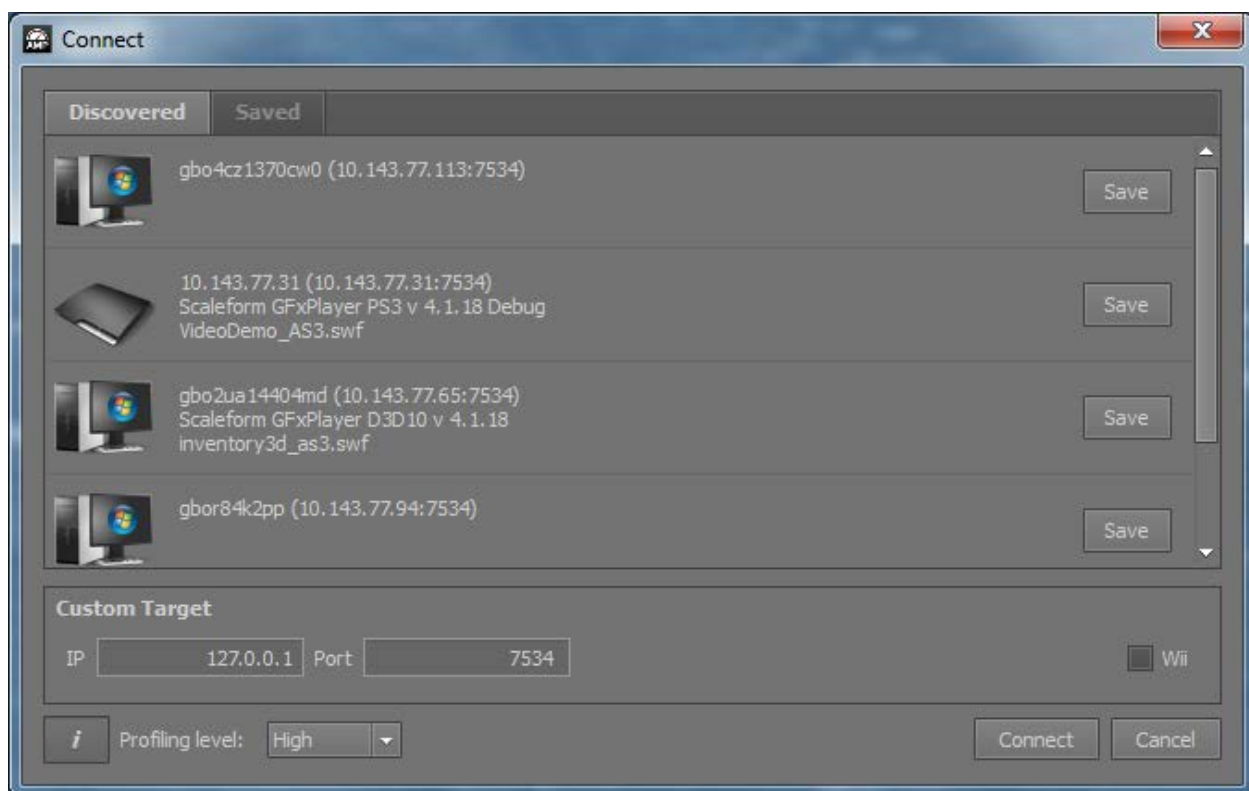
## 2 AMP

AMP                      GfxAmpClient.exe                      .                      Scaleform

*Bin/AMP*                      .

## 2.1

IP . AMP



**2 : AMP**

| File | Connection |
|------|------------|
|------|------------|



CPU CPU Summary, Functions ActionScript  
 , Function  
 , ActionScript

4 : CPU Summary

CPU CPU ,

2.3

f

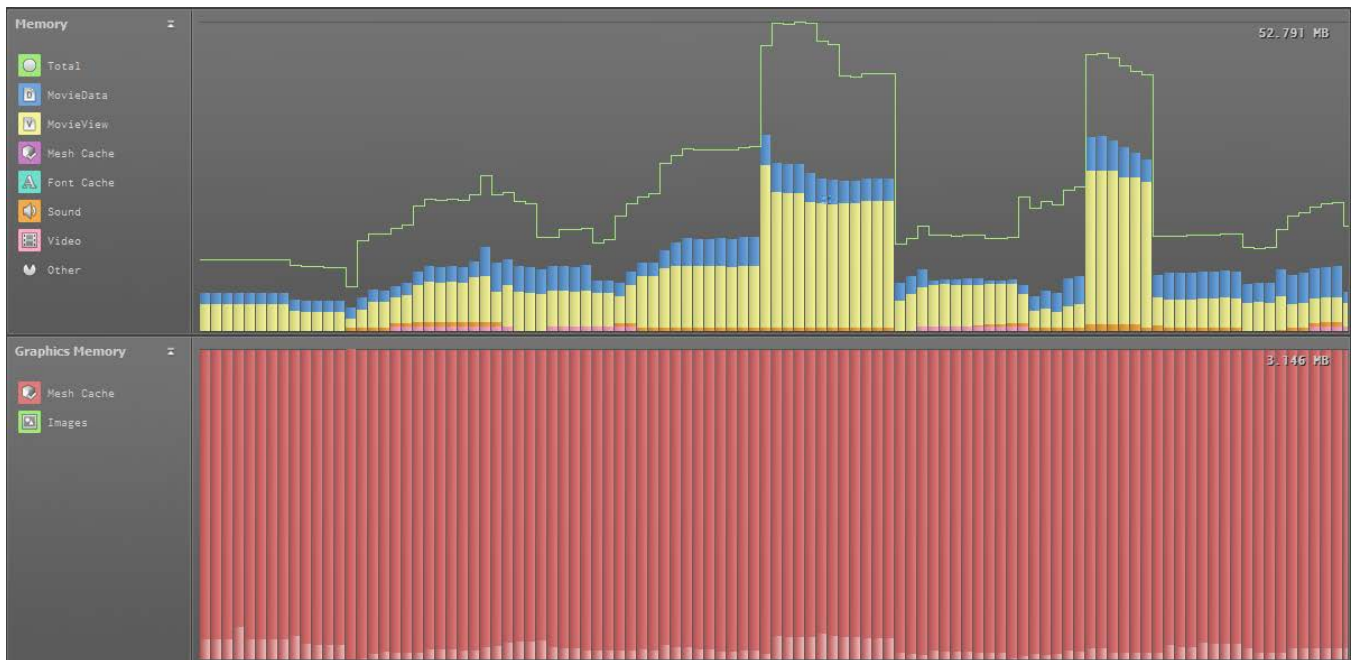
5:

Render Summary

6: Rendering Summary

2.4

Scaleform



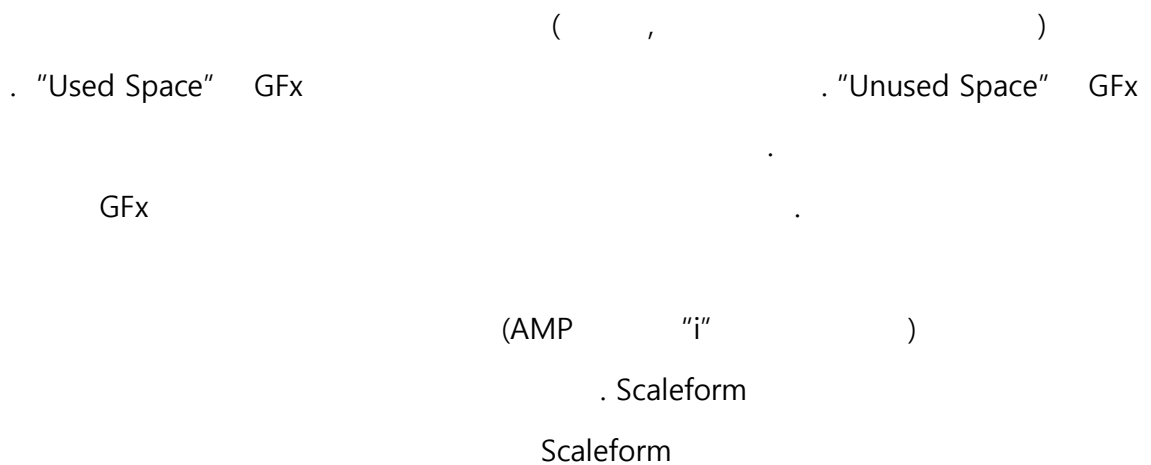
## 7: Memory

- **Total:** Scaleform ( ).  
 ,  
 ,  
 .
- **MovieData:** SWF GFX ,  
 ,  
 .
- **Movie view:** , ActionScript  
 Gfx::Movie .
- **MeshCache:** ,  
 tesselator .  
 , edge antialiasing .
- **Font Cache:** .

- **Sound:**   
- **Video:** 
- **Other:** Scaleform 
- **Images:**  (  ) 

[Memory System Overview](#)

Memory



o **Renderer:** Scaleform 

▪ **Buffers:**

▪ **RenderBatch:** 

- **Primitive:**
- **Fill:**
- **Mesh:**
- **MeshBatch:** (chunk)
- **Context:**
  - (thread)
- **NodeData:**
- **TreeCache:**
- **TextureManager:**
- **MatrixPool:**
- **MatrixPoolHandles:**
- **Text:** , html (line)
- **Font:**
- **MovieDef:**
  - **CharDefs:**
  - **ShapeData:**
  - **Tags:** ActionScript
  - **Fonts:**

- **Sounds:**
- **ASBinaryData:** ActionScript
- **MD\_Other:** MovieDef
- o **MovieView:** ActionScript
  - **MovieClip:**
  - **ActionScript:** ActionScript
    - **ASString:** ActionScript
  - **Text:**
  - **XML:** XML
  - **MV\_Other:** Movie View
  - **VM:** ActionScript
    - **AS3 VM:** ActionScript 3
    - **CallFrame:** ActionScript
    - **VTable:** ActionScript
    - **SlotInfo:** ActionScript
      - ClassTraits    InstanceTraits
      - SlotInfo    ClassTraits
      - InstanceTraits
    - o **SlotInfoHash:** ActionScript (hash)
    - **ClassTraits:** ActionScript
    - **Class:** ActionScript



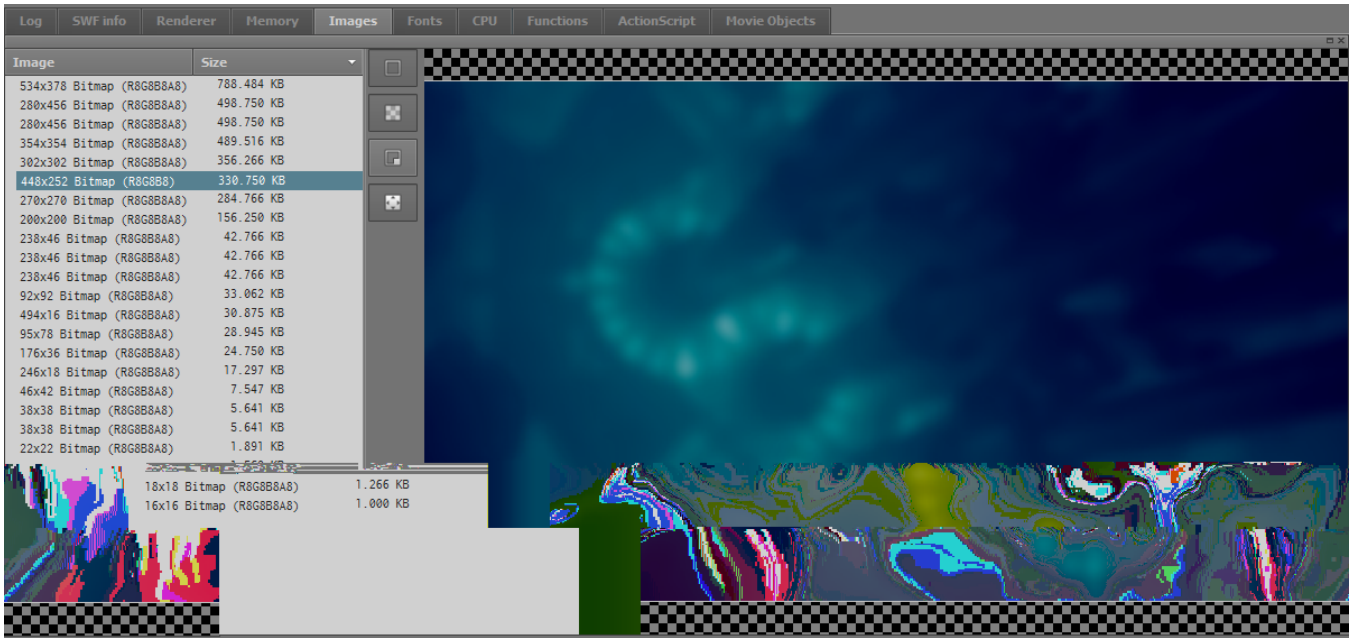
- **InstanceTraits:** ActionScript
- **Instance:** ActionScript
- **AbcFile:**
  - **AbcConstPool:** , , Namespaces Multinames  
(constants)  
AbcFile
- **VMAbcFile:** AbcFiles ,
- **Tracer:**
- o **IME:** Input Method Editor (IME)

| Log    | SWF info                        | Renderer    | Memory | Image |
|--------|---------------------------------|-------------|--------|-------|
| Memory |                                 | Size        |        |       |
| ▲      | Total Footprint                 | 4849.402 KB |        |       |
| ▲      | Used Space                      | 4633.715 KB |        |       |
| ▷      | Video Heaps                     | 0.000 KB    |        |       |
| ▷      | Other Heaps                     | 325.621 KB  |        |       |
| ▲      | Movie View Heaps                | 1226.094 KB |        |       |
| ▲      | MovieView "3dgenerator_as3.swf" | 1226.094 KB |        |       |
|        | String                          | 0.828 KB    |        |       |
| ▷      | Renderer                        | 12.672 KB   |        |       |
| ▷      | MovieView                       | 1176.781 KB |        |       |
|        | General                         | 35.812 KB   |        |       |
| ▷      | Movie Data Heaps                | 261.066 KB  |        |       |
| ▲      | Global Heap                     | 2820.934 KB |        |       |
|        | Video                           | 264.176 KB  |        |       |
|        | String                          | 20.633 KB   |        |       |
|        | Sound                           | 0.141 KB    |        |       |
| ▷      | Renderer                        | 2310.914 KB |        |       |
| ▷      | MovieView                       | 8.039 KB    |        |       |
| ▷      | MovieDef                        | 116.676 KB  |        |       |
|        | Image                           | 0.141 KB    |        |       |
|        | IME                             | 9.914 KB    |        |       |
|        | General                         | 90.301 KB   |        |       |
| ▷      | Debug Memory                    | 8.000 KB    |        |       |
| ▷      | Unused Space                    | 38.820 KB   |        |       |
|        | Debug Data                      | 176.867 KB  |        |       |

## 8: Memory Summary

Scaleform

Images



## 9: Images

Scaleform

..

Scaleform

Fonts



## 10: Fonts

SWF

, MovieData

CharDefs

AMP

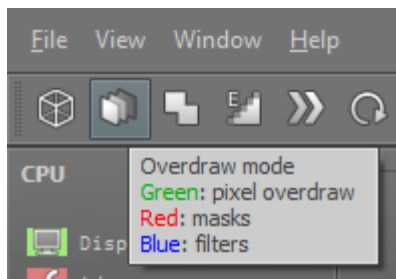
Heaps

Memory Summary

SWF

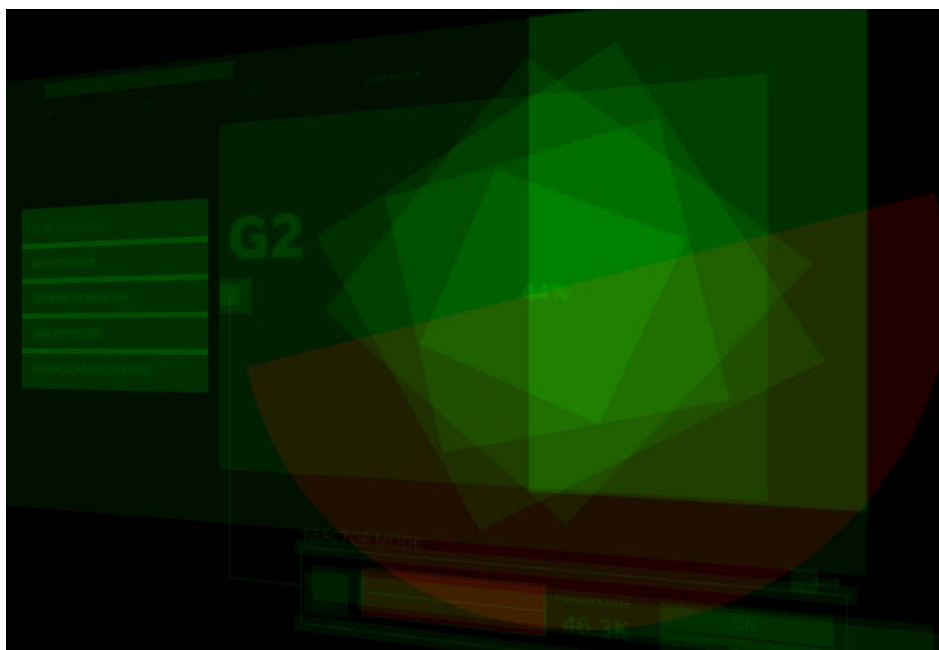
## 2.5

AMP



11:

16 ), , ( 8 ) . ( ,



12: Highlighting

2.6

AMP

Figure 13: Batch profiling mode

### 3

1. AMP                      GfX::System                      GfX::System::Init  
                                  .                      Scaleform::System                      Scaleform                      ,  
                                  GfX::AMP::Server::Init()                      GfX::AMP::Server::Uninit()  
                                  AMP                      .
  
2. AMP                      Windows, Xbox360, PS3, Linux, Mac,Wii, Android                      iPhone/iPad, 3DS, PS  
                                  Vita, WiiU                      . AMP  
                                  .                      AMP                      IP                      ,                      IP  
                                  .  
                                  • Xbox360                      IP                      IP                      .                      IP  
                                                       Xbox 360 Neighborhood                      .  
                                  • PS3                      IP                      IP  
                                                       IP                      .  
                                  • Android                      .  
                                                       AndroidManifest.xml                      .  
                                                       <uses-permission android:name="android.permission.INTERNET" />  
                                  • WiFi                      PC                      WiiU                      .                      PC  
                                                       6003                      .                      TCP/IP  
                                  .  
                                  • Wii                      AMP                      .                      NDEV  
                                                       USB                      (COM                      )                      PC                      . IP  
                                                       "wii"                      Wii AMP                      .  
                                                       : RVL\_SDK/X86/bin                      .                      hio2.dll  
                                  AMP                      .
  
3. Unreal Engine 3                      AMP                      Flash  
                                  SWD/AS3 AS                      .                      AMP                      SWF

AS2 SWD

SWD/AS

AMP

.

## 4

1. AMP non-shipping Scaleform .  
 Scaleform AMP GfxConfig.h SF\_AMP\_SERVER Gfx  
 . Scaleform  
 AMP . Ptr<Gfx::AMP::ServerState> serverState = \*SF\_NEW  
 Gfx::AMP::ServerState();  
 serverState->StateFlags &= Gfx::AMP::Amp\_Disabled;  
 AmpServer::GetInstance().UpdateState(serverState);
2. AMP .  
 (Windows: Ws2\_32.lib,  
 Xbox 360: Xnet.lib, PS3: libnetctl, Linux: ).
3. AMP , AMP  
 .  
 .  
 .  
 , AMP  
 .  
 AMP  
 . AMP::Server::GetInstance().SetInitSocketLib(false).  
 AMP .
4. AMP . Scaleform  
 .  
 AMP::Server::GetInstance().SetHeapLimit() .
5. AMP 7534 .  
 . AMP::Server::GetInstance().SetListeningPort();



## 6. AMP

```
. AMP::Server::GetInstance().SetConnectionWaitTime.
```

## 5 AMP

Scaleform

AMP

AMP

Scaleform Player

### 5.1 Debug

AMP    ActionScript 2 Flash    (SWD    )

    . Flash CS3    CS4    (Ctrl+Shift+Enter)    SWD

    . SWD    SWF    AMP

ActionScript 3    SWD    SWF

    AS    . Flash Studio    SWF

### 5.2

AMP

AMP

AMP

- AMP::AppControlInterface

, AMP

AMP::MessageAppControl

HandleAmpRequest

- AMP::Server::GetInstance().SetAppControlCallback  
.
- AMP::Server::GetInstance().SetAppControlCaps AMP  
.  
AMP::MessageAppControl , true  
:  
  
AMP::MessageAppControl caps;  
caps.SetCurveToleranceDown(true);  
caps.SetCurveToleranceUp(true);  
caps.SetNextFont(true);  
caps.SetRestartMovie(true);  
caps.SetToggleAaMode(true);  
caps.SetToggleAmpRecording(true);  
caps.SetToggleFastForward(true);  
caps.SetToggleInstructionProfile(true);  
caps.SetToggleOverdraw(true);  
caps.SetToggleBatch(true);  
caps.SetToggleStrokeType(true);  
caps.SetTogglePause(true);  
caps.SetToggleWireframe(true);  
AMP::Server::GetInstance().SetAppControlCaps(&caps);

### 5.3

AMP

AMP::Server::GetInstance().SetConnectedApp

AMP

- ServerStateType  
AMP::Server::GetInstance().SetState.  
o Amp\_RenderOverdraw

- o Amp\_App\_Wireframe
- o Amp\_App\_Paused
- o Amp\_App\_FastForward
- AMP::Server::GetInstance().SetAaMode
- AMP::Server::GetInstance().SetStrokeType
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::ServerState
- AMP::Server::GetInstance().UpdateState

## 5.4

AMP C++ ActionScript CPU  
 . ActionScript

Amp.addMarker(1)

: Amp.addMarker() .as

Resources/AS2/CLIK

Amp

C++

Gfx::MovieImpl:: AdvanceStats->AddMarker(1)

AMP

# 6 AMP FAQ

AMP .

## 1. AMP ?

GFX\_AMP\_SERVER AMP . Gfx

AMP . Platform::RenderThread::drawFrame  
AmpServer::GetInstance().AdvanceFrame AMP

## 2. AmpClient .

AmpServer::AdvanceFrame() ?

AMP .  
AMP

.  
.  
.  
AMP  
"I"( I) .

## 3. Windows/PS3/Xbox . AMP AMP

AMP .  
Platform::RenderThread::drawFrame AmpServer::GetInstance().AdvanceFrame  
AMP .

## 4. (AMP::Server::WaitForAmpConnection ) AMP ?

AMP

AMP

Scaleform

AMP

'i'

. AMP

'profile level'

1MB

GFx::AMP::Server::SetHeapLimit

**5.**

**?**

AMP

1MB

AmpServer::SetHeapLimit

AMP

GFx

# 7

AMP

- [AMP Use Cases](#) .
- [Scaleform Reference](#) – – /
- [AMP Forum](#) –